MAPEX: Modality-Aware Pruning of Experts for Remote Sensing Foundation Models

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***Abstract*—Remote sensing data is commonly used in a wide range of tasks, such as natural disaster monitoring and land-use studies. For each task, scientists carefully choose appropriate modalities or leverage data from purpose-built instruments. Recent work on remote sensing foundation models pretrains computer vision models on large amounts of remote sensing data to learn general-purpose representations. However, this progress comes at the cost of very large models, which are particularly challenging to deploy on edge devices due to their high inference costs. Moreover, downstream applications often rely on only a subset of modalities and operate under strict resource constraints, creating a mismatch between the large, multipurpose models produced by pretraining and the lightweight, task-specific models needed in practice. We address this mismatch with modality-aware pruning of experts (MAPEX), a remote sensing foundation model based on mixture-of-modality experts. MAPEX is pretrained on multimodal remote sensing data using a novel modality-conditioned token routing mechanism that naturally encourages the emergence of modality-specialized experts. To apply the model on a specific task, we propose a modality-aware pruning technique that retains only the experts relevant to the task's modalities, resulting in lightweight, task-specific models that can be directly extracted from the pretrained foundation model, at no additional cost. Our approach yields efficient modality-specific models while simplifying fine-tuning and deployment for the modalities of interest. We experimentally validate MAPEX on diverse remote sensing datasets and show strong performance compared with fully supervised training and state-of-the-art remote sensing foundation models. Code will be available at github.com/HSG-AIML/MAPEX**

***Index Terms*—Foundation model, mixture-of-expert, multimodality.**

I. Introduction

COMPUTER vision methods are widely applied in the remote sensing domain for tasks such as classification, segmentation, or detection in imagery obtained from satellites or aerial platforms [1]. Increasingly, the visual foundation models tailored to characteristics of remote sensing imagery, such as top-down viewpoint, varying resolution, or multispectral data, are adopted for these tasks [2], [3].

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Advanced foundation models pretrained on unlabeled data with self-supervised objectives present a great opportunity in the remote sensing domain, where labeled datasets are small and expensive, but unlabeled data is widely available in huge quantities [4]. This is particularly important for tasks that rely on scarce annotations, where labels are costly to obtain and only weak supervision is often available [5].

However, existing foundation models for remote sensing still face two fundamental mismatches with real-world applications. The first is a modality mismatch: most remote sensing foundation models are pretrained on individual modalities such as optical RGB [6] or multispectral [7], [8]. This becomes challenging when the downstream task requires a different modality, for example, applying a model trained on multispectral data to radar-based applications [6], [8]. In contrast, multimodal foundation models attempt to overcome this by representing all modalities jointly [9], [10], [11], but this creates a size mismatch: such models are typically very large, which is advantageous during pretraining but makes them costly to store, fine-tune, and deploy on edge devices, in particular when only a subset of modalities is needed downstream [11], [12], [13].

Models that exclusively work with multiple modalities can be restrictive in remote sensing, where instruments and satellite missions are often designed to solve specific tasks, such that multimodal data might not be necessary downstream. For instance, synthetic aperture radar (SAR) data is commonly used to map flooding because it is sensitive to water cover and able to penetrate clouds, which are common around flood events [14]. Pairing SAR with optical data that cannot penetrate cloud cover for the sake of maintaining the multimodal pretraining setup for downstream fine-tuning (FT) might not be useful in such cases. Similar arguments could be made for a series of other remote sensing tasks in areas ranging from wildfire detection [15] to environmental monitoring or agricultural applications [16].

While multimodal pretraining gives computer vision models a more complete picture of the world and allows them to leverage valuable synergies between instruments [17], there remains a clear gap between the large foundation models produced by pretraining and the lightweight, task-specific models needed in practice. We address these mismatches by introducing a novel modality-aware pretraining method and successive expert-pruning step. First, we propose a novel pretraining method for multimodal remote sensing data based on a mixture-of-experts (MoEs) architecture with modality-

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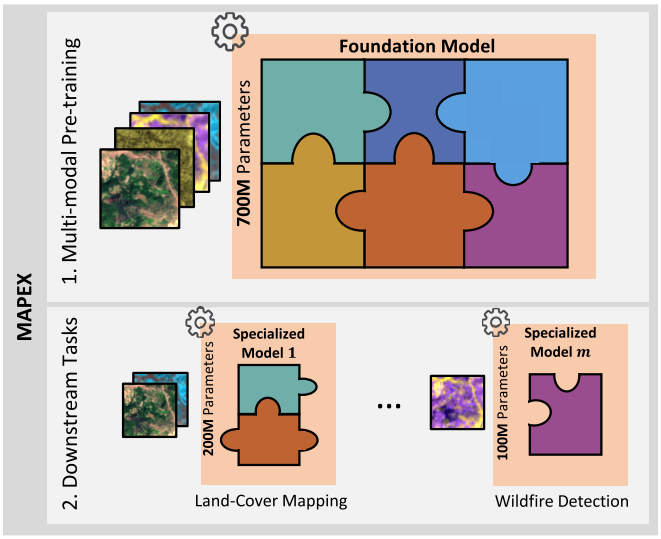


Fig. 1. Overview of MAPEX. (Top) Multimodal foundation model is pretrained using a MoE architecture with modality-aware routing. (Bottom) During FT, irrelevant experts are pruned, yielding smaller, specialized models tailored to specific downstream tasks (e.g., land-cover mapping and wildfire detection), making them lighter and easier to fine-tune.

conditioned token routing. Unlike standard MoE approaches that route tokens individually based on feature similarity, MAPEX performs modality-level routing, where all tokens from the same modality share the same expert selection determined by learned modality embeddings. This method facilitates the use of multimodal pretraining data while enforcing modality-specific structure in large parts of the pretrained model. In the second step, we leverage this modality-specific structure with modality-aware pruning of experts (MAPEX) for the modality of the downstream task (see Fig. 1). This yields a significantly smaller, more focused model that is specialized for one (or multiple) modalities of interest and can be easily fine-tuned on that modality. By doing so, MAPEX can adapt to any set of available modalities during inference (solving the modality mismatch) while producing much smaller, task-specific models compared with the original foundation network (solving the size mismatch). This approach allows us to combine the best of both worlds: leverage large-scale multimodal data for self-supervised pretraining of large foundation models, but also provide small, specialized models that are easy to use and pretrained for a wide variety of modalities.

In summary, our contributions are as follows.

1) A new multimodal pretraining approach for a remote sensing MoEs model, enabling a single large foundation model to learn modality-specialized experts.
2) A novel modality-conditioned expert routing method to create modality-specific experts within the multimodal model.
3) A modality-aware pruning technique that transforms this large foundation model into multiple lightweight, modality-specific models tailored to different downstream tasks.

II. Related Work

A. Foundation Models

Foundation models are based on the idea of training big models on large, unlabeled data with self-supervised learning techniques before adapting them to specific downstream tasks [18]. This approach has produced strong, general models for natural language [19], computer vision [20], and more specialized variants for domains such as remote sensing [21]. Increasingly, research in foundation models has focused on multimodal data, particularly the combination of language and visual data [22]. Multimodal foundation models benefit from training on datasets of multiple paired modalities, producing models that perform well on individual modalities and enabling novel applications with multimodal data (see [23] for a review). Large-scale pretraining improves performance on downstream applications, but also requires big models for optimally compute-efficient training [24]. This increases latency as well as deployment and model serving costs, limiting foundation model use in time or cost-sensitive settings. A growing body of literature addresses the foundation model efficiency issues with techniques like pruning [25], distillation [26], or parameter-efficient FT methods [27].

B. Mixture-of-Experts

A complementary area of research focuses on conditional computation as a way of increasing model size efficiently. MoEs models are commonly used in large foundation models [28], [29], [30]. These models consist of multiple expert modules and a router that assigns tokens to each expert during training and inference. In standard MoE models, the router r produces logits from tokens t as $r(t) = W_r \cdot t$, which are turned into routing probabilities with a softmax to determine the routing of t. In transformer models, MoEs replace feed-forward network (FFN) layers with multiple experts, each implemented as an FFN itself. This approach has been used to train performant large-scale language [31] and vision models [29] and has recently gained traction in remote sensing [32], [33], [34]. Sparse MoE models conditionally route tokens to a subset of expert modules, making inference more efficient [31], while large-scale distributed training benefits from conditional computation by facilitating parallelism across GPUs [28].

In the multimodal context, MoEs have been applied to language-image models [35], incorporating modality-specific experts with token routing based on each token's modality. Furthermore, dataset-aware mixture-of-experts (DAMEXs) [36] introduced a dataset-aware MoE model for object detection where experts specialize in specific datasets rather than modalities. In DAMEX, the MoE layers process data from different object detection datasets with experts specifically dedicated to each dataset. In contrast to our work, the model is trained in a supervised fashion for object detection with mostly optical RGB data and without an explicit pruning step for inference or FT on individual datasets.

C. Remote Sensing Foundation Models

Remote sensing enables Earth observation through active and passive sensors mounted on satellites or aerial platforms,

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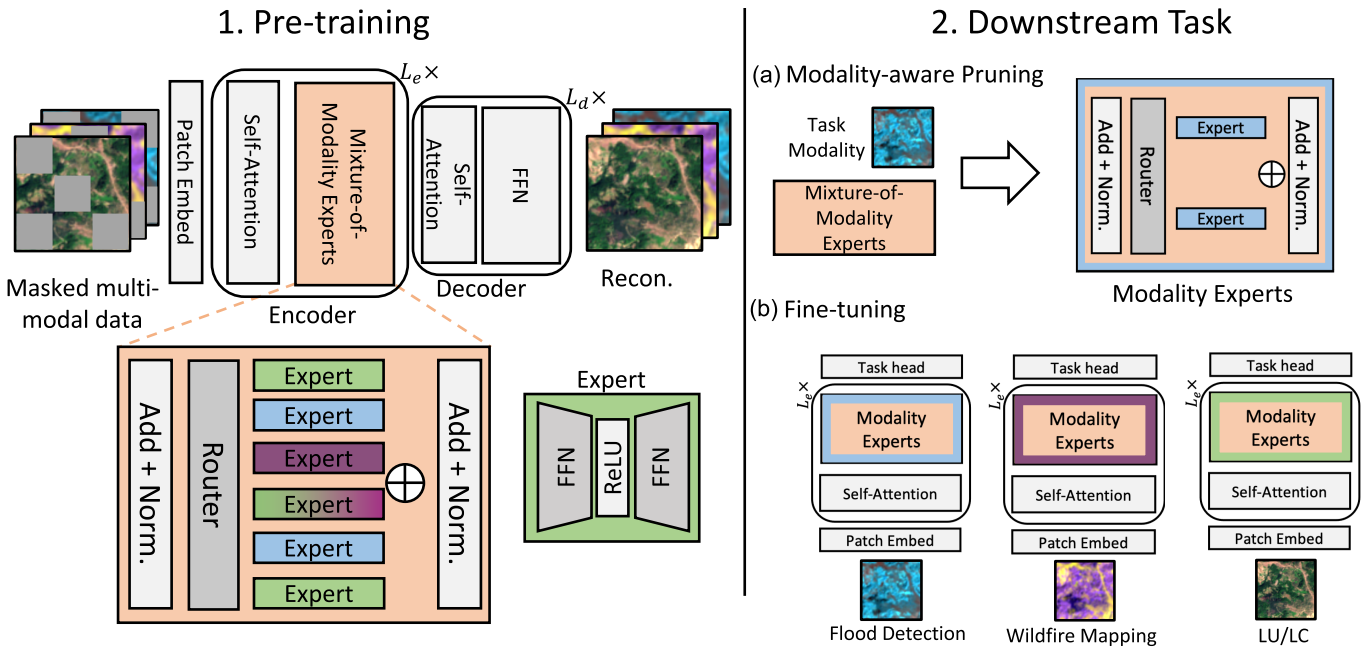


Fig. 2. Overview of the pretraining and downstream application process: 1) Model is trained with all modalities on a masked reconstruction task. Modalities are processed by specialized experts in the mixture-of-modality experts module: 2a) Modality experts are pruned for a modality of interest, and 2b) the resulting modality experts model is fine-tuned on a downstream task of the corresponding modality.

capturing vast amounts of heterogeneous data [37], [38]. Unlike traditional RGB imagery, satellite data varies significantly across sensor types, making model transferability challenging. To address this, foundation models for remote sensing leverage large-scale self-supervised pretraining on unlabeled data. Most foundation models target specific modalities, such as optical RGB [8], [6], multispectral [21] images which sample the electromagnetic spectrum with a few broad bands, hyperspectral [39] data, which provides almost continuous coverage of the electromagnetic spectrum with many narrow bands, or SAR [40] images that are acquired through active sensing with radar waves. More recent works, such as MMEarth [12], DOFA [9], RingMo-SAM [11], SkySense [13], and SSL-based multimodal models [41], explore large-scale multimodal pretraining by integrating diverse sensor data to improve generalization. However, these models typically remain monolithic, processing all available modalities. In contrast, MAPEX introduces a selective MoEs approach that activates only the relevant experts per input, enabling the extraction of smaller, task-specific models for downstream applications. This improves computational efficiency while preserving strong multimodal performance.

III. Methods

Fig. 2 presents a high-level overview of our method.

A. Problem Setup

We consider a multimodal dataset $\mathcal{D}_n^m$ consisting of n co-located and temporally matched samples from m different modalities. Our model is based on the Vision Transformer (ViT) base architecture [42]. Input samples $\mathbf{x}_i^j \sim \mathcal{D}_n^m$ of shape $\mathbf{x} \in \mathbb{R}^{\left(\sum_j^m C^j\right) \times H \times W}$ are split into patches and linearly embedded per modality with learnable patch-embedding transforms $\{f^j\}_{j=1,\ldots,m}$. This yields a set of m D-dimensional tokens for each sample patch. Sinusoidal positional embeddings to encode spatial information are added to each token [43]. We investigate sinusoidal and learnable modality-tokens to encode the modality of tokens across the channel dimension. To mark modality transitions in the token sequence, learnable end-of-modality tokens are injected before attention operations.

B. Masked Image Modeling for Multimodal Pretraining

We utilize masked image modeling as a self-supervised pretraining technique [44]. After patch-embedding, a binary mask is applied to the input sequence of tokens, independently masking 75% of the tokens corresponding to each modality as in [44]. The remaining tokens are processed by a ViT encoder. Once encoded, the original sequence length is restored by introducing special, learnable, `[MASK]` tokens at the positions of dropped input tokens. A ViT decoder then aims to reconstruct the original (normalized) pixel values of the masked patches. We use a mean-squared error (mse) objective between pixel values of the original masked patches and their reconstruction for training [44].

1) Modalitywise Decoding: To keep the reconstruction task difficult in the presence of multiple independently masked modalities (a spatial patch might be masked in one modality but visible in another), we reconstruct all modalities independently. All tokens of a modality are processed by the decoder in a distinct forward pass, incurring a total of m decoder forward passes, while also reducing the sequence length in each pass by a factor of $1/m$.

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C. Mixture-of-Modality Experts

MoEs models replace the feed-forward layers in each ViT block with a routing module and multiple feed-forward expert modules [28]. The router consists of a linear layer that produces logits for each token, from which a softmax distribution over the e experts is constructed. The expert outputs for each token are then weighted by the routing softmax weights. Unlike token-based routing, MAPEX performs modality-level routing, where all tokens from the same modality are routed to the same experts. We adapt the MoE framework to enforce expert ↔ modality relationships in the model. Starting from a fixed assignment of experts to modalities, we ultimately design a learnable routing mechanism based on modality information that consistently routes tokens from a modality M to the same subset of experts. Top-k selection of the k highest ranked experts is used to achieve sparse conditional computation for each modality. Our deterministic routing setup defines a fixed one-to-one relationship between modalities and experts. Every modality is assigned to exactly one expert a priori, and no experts are shared between modalities. Positional embedding routing utilizes sinusoidal positional embeddings [43] across the stack of modalities as input to the routing module. While the routing weights are learned, the input embeddings remain fixed and unique for each modality. This approach supports an arbitrary number of experts but introduces an arbitrary ordering among modalities. Modality routing introduces learnable [MODALITY] embeddings that represent each modality in lieu of channelwise positional embeddings. Unlike tokens based on modalities' relative position in the input data, learnable embeddings do not introduce an implicit order between the modalities. These tokens are added to each modality and used as input to the routing module. Explicitly, for token t^m of modality m, the routing weights are computed by passing the associated modality embedding to the routing module r: $w_r^m = r([\texttt{MODALITY}]^m)$. Unlike standard MoE implementations, this routing strategy guarantees that tokens of a modality will always be routed to the same experts, even across samples.

1) Load-Balancing Loss: Imbalanced expert utilization is a common problem with learned routing mechanisms [28]. We observe that some experts randomly receive a lower share of tokens at the beginning of training, resulting in fewer gradient updates and continued low utilization of these experts throughout training. We address this issue by introducing a differentiable load-balancing loss that encourages a more uniform distribution of routing probabilities between experts

$$\mathcal{L}_{\text{load}} = \frac{1}{e}\sum_{i=1}^{e}\left(U_i - \frac{1}{e}\right)^2 \tag{1}$$

with e the total number of experts and U_i the empirical utilization fraction of the ith expert (i.e., the fraction of all tokens that expert i processes). This additional loss term encourages more uniform expert utilization. During pretraining, the load-balancing loss is combined with the reconstruction loss to create the training objective: $\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha \cdot \mathcal{L}_{\text{load}}$ where the weighting factor α scales the load-balancing loss.

2) Shared Expert: We hypothesize that some features are shared between modalities and introduce an additional shared expert that gets access to tokens from all modalities. The shared expert can process common features (such as edges, corners, or colors) that are shared between modalities, allowing the modality experts to focus on information that is specific to individual modalities. We limit the number of shared experts to one to maintain lightweight modality-specific models after pruning (see Section V-A4 for a discussion on the tradeoff between model size and number of experts).

3) Modality Dropout: In order to prime the model for downstream applications where the full set of pretraining modalities is not available, we utilize modality dropout during the pretraining stage. In each batch, the tokens corresponding to a predefined fraction of modalities are set to zero.

D. Downstream Modality-Aware Expert Pruning

After pretraining the mixture-of-modality experts model with masked autoencoding on unlabeled multimodal data, we transfer the model encoder to the downstream task of interest. Often, only a subset of the pretraining modalities will be available for the downstream task. Accordingly, we prune the mixture-of-modality experts to suit the available data modalities. For a model pretrained with experts E^p on $\mathcal{M}^p$ pretraining modalities and k experts per modality, all experts and patch-embedding projections corresponding to modalities that are not in the set of downstream modalities are dropped. The downstream experts E^d are extracted from each pretrained model layer as

$$E^d = \left\{E_i \,\middle|\, i \in \arg\operatorname{top} k\left(w_r^m\right) \forall m \in \mathcal{M}^d\right\}_{i=1,\ldots,|E^p|} \tag{2}$$

with w_r^m the pretraining routing probabilities for modality m. In the case of a single downstream modality, this yields a model with k downstream experts. To utilize the pruned model on a supervised downstream task, we perform standard FT on labeled data. This step adapts the purely unsupervised multimodal model trained for reconstruction to modality-specific supervised tasks.

IV. Datasets

A. BigEarthNet-GeoEnvironment (ben-ge)

The *ben-ge* dataset [45] is an extension to the BigEarthNet-MM [46] dataset. It consists of 590 326 co-located and temporally matched samples from Sentinel-1 (SAR), Sentinel-2 (multispectral), as well as elevation and environmental data. We utilize the *ben-ge* dataset for multimodal pretraining of the mixture-of-modality experts model. We group the data into six distinct modalities: RGB (Sentinel-2 B04, B03, B02), red edge (Sentinel-2 B05, B06, B07), short-wave infrared (SWIR, Sentinel-2 B11, B12, B8A), near-infrared (NIR, Sentinel-2 B08), *SAR* (Sentinel-1 VV and VH polarizations), and *ELE* (elevation map). *ben-ge-8k* is a subset of *ben-ge* consisting of ≈8000 samples that we use for downstream evaluation on multilabel land-use/land-cover classification with different modalities.

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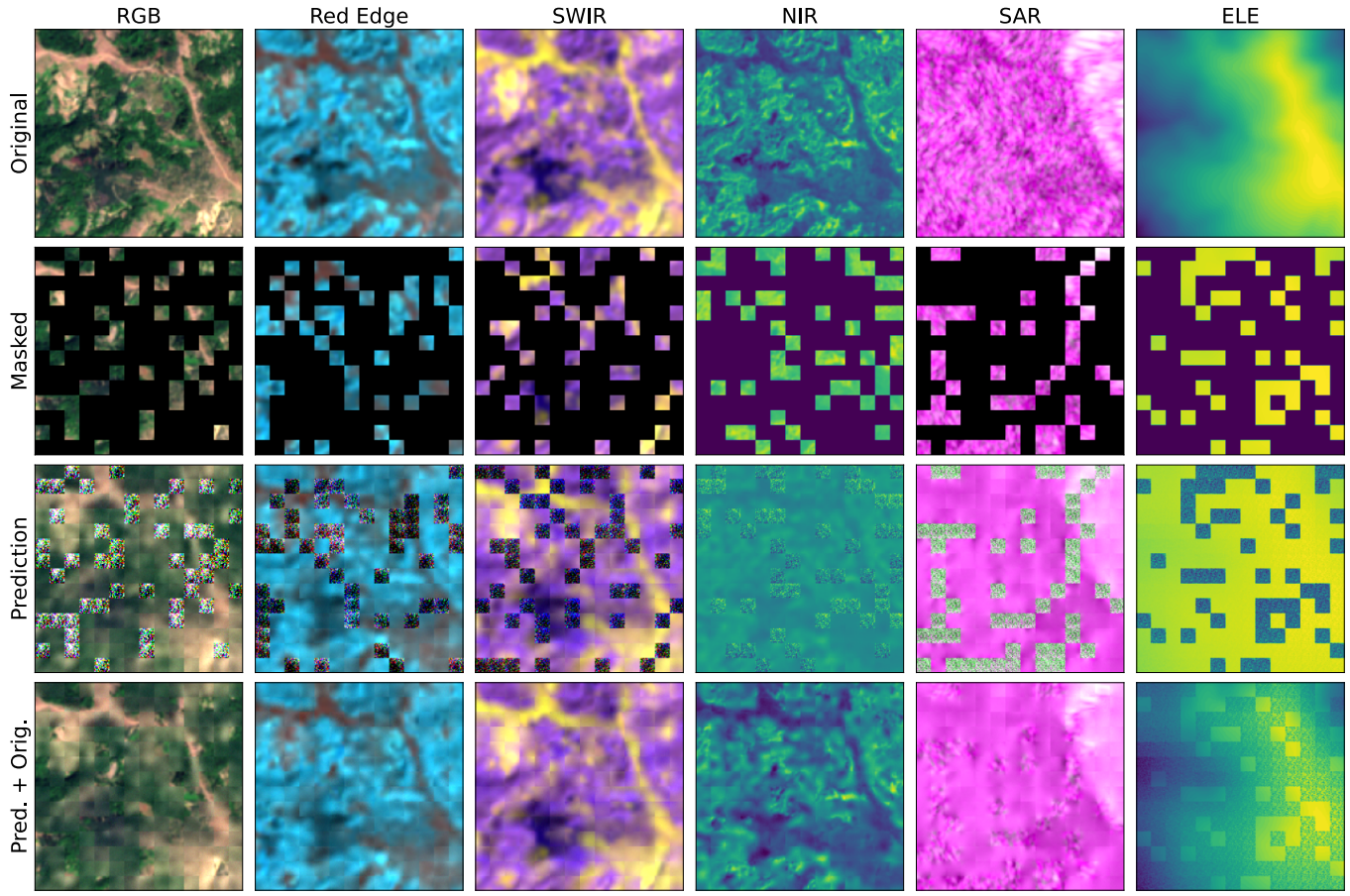


Fig. 3. Reconstruction examples from mixture-of-modality experts model on multimodal data. (Left to right) Optical RGB, red edge, SWIR, NIR, SAR, and elevation from *ben-ge* [45]. The same decoder is used for all modalities.

B. SEN12-FLOOD

This dataset consists of Sentinel-1 and Sentinel-2 imagery acquired in areas prone to flooding and corresponding binary flood labels [47] across the world. It contains 366 time-series of varying length, with 512×512 pixels per tile. The dataset is split by locations, and tiles within a time series are considered independently. We use the SAR data from SEN12-FLOOD for downstream evaluation of the pretrained model on a single modality.

C. California Wildfire

We construct a dataset specifically for wildfire detection by collecting fire perimeter data from the California Fire Perimeters dataset [48] and recent CAL FIRE incident records [49]. These records include detailed spatial and temporal information on wildfire events, such as locations, dates, and sizes. For each fire perimeter, we acquire corresponding Sentinel-2 satellite imagery [50]. To create a binary classification dataset, we crop small image patches (224×224 pixels) and label them as either `wildfire` or `no-wildfire` based on overlap with the wildfire perimeter. We utilize Sentinel-2 SWIR bands (B8A, B11, and B12), which have proven most effective in detecting burned areas [15], [51]. The resulting dataset contains 20 000 samples, with 11 000 labeled as `wildfire` and 9000 as `no-wildfire`.

V. Experiments and Results

A. Self-Supervised Pretraining

MAPEX models are pretrained with masked autoencoding on multimodal remote sensing data with balanced modality shares. Representations generated by the mixture-of-modality experts encoder (six experts, 360M parameters) are mapped back to pixel-space for each modality with an eight-block transformer decoder (see Section III-B). We pretrain for 50 epochs with AdamW [52] on the *ben-ge* dataset. The load-balancing loss is weighted by $\alpha = 0.01$, which yields near uniform expert utilization. Pretraining results in good reconstruction performance (see Fig. 3) for most modalities. Reconstruction is most difficult for the SAR backscatter data, where reconstruction mse remains $\approx 3\times$ higher than on the other modalities.

1) Routing Mechanism: In contrast to standard MoE approaches, this work aims to elicit strong expert $\leftrightarrow$ modality relationships during pretraining, which facilitates expert-pruning for downstream tasks. In standard token-based routing, routing probabilities for each token t are computed by the

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TABLE I

COMPARISON OF ROUTING MECHANISMS USING k-NN ACCURACY FOR LAND-COVER CLASSIFICATION ON DIFFERENT *ben-ge-8k* [45] MODALITIES

Modality	Deterministic	Pos. Embed	Modality-based
RGB	63.5	63.1	**67.2**
Red Edge	63.2	64.9	**66.7**
SWIR	68.7	67.9	**72.4**
NIR	51.6	55.9	**56.8**
SAR	62.2	65.3	**66.1**
ELE	46.0	**48.3**	48.2

TABLE II

EFFECT OF UNI VERSUS MULTIMODAL PRETRAINING AND NUMBER OF PRETRAINING EXPERTS ON *ben-ge-8k* [45]. k-NN ACCURACY ON THE RGB MODALITY AFTER PRUNING ALL MODELS TO ONE EXPERT

Pre-training Modalities	Pre-training Experts	k-NN Acc.
1 (RGB)	1	62.1
6 (All)	1	63.2
6 (All)	6	65.1

TABLE III

k-NN PERFORMANCE FOR LAND-COVER CLASSIFICATION ON DIFFERENT MODALITIES OF THE *ben-ge-8k* DATASET [45], WITH AND WITHOUT THE USE OF A SHARED EXPERT

Modality	2 Experts	2 Experts + 1 shared
# params	130M	200M
RGB	67.2	**67.6**
Red Edge	66.7	**67.1**
SWIR	72.4	**72.9**
NIR	**56.8**	56.0
SAR	66.1	**66.7**
ELE	**48.2**	47.8

router as $w_r = r(t)$. Initial experiments with token-based routing on multimodal remote sensing data failed to produce a clear specialization of experts for individual modalities and suffered from unbalanced expert utilization. We address and evaluate these shortcomings with three alternative routing mechanisms (see Section III-C): deterministic routing, positional embedding routing, and modality-based routing. We find that pretraining with modality-based routing consistently outperforms the two other routing mechanisms across downstream tasks of different modalities (see Table I). This confirms our hypothesis that the routing parameters should be learnable (unlike the deterministic routing), and that learnable modality tokens are a suitable input for the routing module.

2) Multimodal Pretraining: We assess the effect of multimodal pretraining on the *ben-ge* dataset. We compare uni-modal pretraining (only on RGB) against multimodal pretraining (using all available modalities). After pretraining, all models are pruned to a single expert. Table II (first two rows) reports the k-NN accuracy on the RGB modality. The results show that multimodal pretraining leads to better performance on the RGB downstream task compared with pretraining exclusively on the RGB modality. This suggests that the model benefits from shared representations learned across multiple modalities, even when only RGB is available at test time.

3) Expert Modules: To evaluate the benefit of using multiple expert modules in multimodal data, we compare multimodal MAPEX variants with a single and multiple experts (see Table II; last two rows). After pretraining, both models are pruned to one expert and evaluated on the RGB modality. The model pretrained with multiple experts significantly outperforms the model with one expert. Please note that both models are the same size after the pruning stage.

4) Shared Expert: We study the impact of using a shared expert on MAPEX performance across multiple modalities. The shared expert is hypothesized to capture common cross-modal information, enabling modality-specific experts to specialize in unique features. We fine-tune the shared expert along with the retained modality-specific experts. Table III indicates that, while the shared expert generally enhances performance across most modalities, it also introduces a significant increase in model parameters. This raises a tradeoff between performance improvement and computational efficiency, suggesting that the shared expert may not always provide a cost-effective benefit, particularly in resource-constrained applications.

5) Modality Dropout: We investigate the impact of varying modality dropout values (0%, 10%, and 50%) during pretraining on downstream model performance. Table IV shows that a dropout value of 50% outperforms both 10% and 0% by around 1.5% (absolute) on average. Intuitively, a higher dropout value encourages the model to become more robust by learning representations that do not rely on the presence of all modalities. By setting a modality dropout value of 50% during pretraining, the model learns to handle partial data input, which aligns better with scenarios where some modalities may be unavailable. This approach is particularly beneficial as a significant portion of the model is shared during pretraining, with only certain parts pruned for downstream tasks. As a result, the shared component gains robustness from the dropout, enhancing the model's adaptability across different downstream specialized tasks.

TABLE IV

COMPARISON OF DIFFERENT MODALITY DROPOUT VALUES. k-NN ACCURACY FOR LAND-COVER CLASSIFICATION ON DIFFERENT MODALITIES OF THE *ben-ge-8k* DATASET [45] IS REPORTED

Modality	Dropout = 0	Dropout = 0.1	Dropout = 0.5
RGB	65.8	65.8	**67.2**
Red Edge	66.1	66.2	**66.7**
SWIR	71.6	71.9	**72.4**
NIR	54.2	54.3	**56.8**
SAR	65.7	65.8	**66.1**
ELE	46.6	47.1	**48.2**

B. Downstream Evaluation

We evaluate pretrained MAPEX across several downstream tasks and compare it to other MAE-based remote sensing foundation models. MAPEX is based on the ViT-base architecture and retains two experts after pruning for one modality. SatMAE [8] is a ViT-base model trained on multispectral

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TABLE V

k-NN AND FT PERFORMANCE (ACCURACY IN %) FOR LAND-COVER CLASSIFICATION ON DIFFERENT MODALITIES OF THE *ben-ge-8k* DATASET [45]. FOR EACH MODALITY, THE BEST PERFORMING MODEL IS HIGHLIGHTED IN BOLD

Modality	Supervised FS	SatMAE [8]		Scale-MAE [6]		MAPEX (Ours)	
		k-NN	FT	*k*-NN	FT	*k*-NN	FT
Architecture # params	*MAPEX* 130M	*vit_base* 90M		*vit_large* 310M		*MAPEX* 130M	
RGB	71.2	64.5	70.8	75.1	**79.9**	67.2	75.6
Red Edge	73.3	65.5	69.7	74.7	**80.3**	66.7	77.8
SWIR	76.5	68.3	74.5	76.2	**83.7**	72.4	79.8
NIR	56.1	50.0	56.9	49.9	57.2	56.8	**59.4**
SAR	69.7	62.2	67.2	65.3	73.1	66.1	**73.6**
ELE	51.8	46.5	47.1	46.6	51.4	48.2	**58.8**

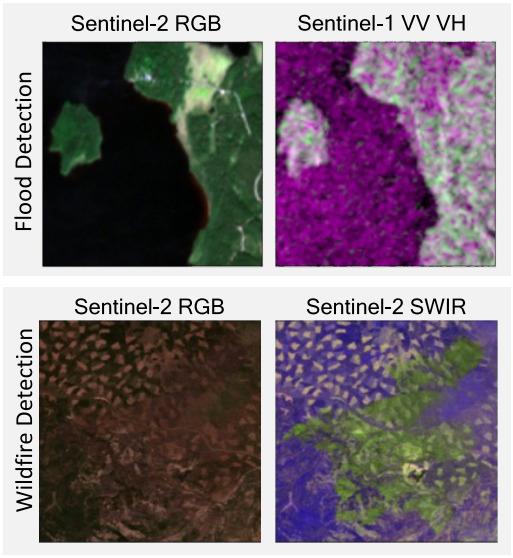


Fig. 4. Task-specific modalities: Sentinel-1 bands provide valuable information for detecting water bodies in the presence of clouds. Similarly, SWIR bands are more effective than RGB for identifying burned areas.

TABLE VI

k-NN AND FT PERFORMANCE (ACCURACY IN %) FOR BINARY FLOOD DETECTION USING SENTINEL-1 DATA [47] AND WILDFIRE DETECTION [48] USING SWIR DATA (SENTINEL-2 B8A, B11, B12 BANDS)

Modality	FS	SatMAE		Scale-MAE		MAPEX	
		k-NN	FT	*k*-NN	FT	*k*-NN	FT
Architecture # params	*MAPEX* 130M	*vit_base* 90M		*vit_large* 310M		*MAPEX* 130M	
Flood Detection	78.1	77.4	80.3	83.0	82.6	80.1	**83.2**
Wildfire Detection	84.7	84.2	88.6	86.6	**90.8**	87.3	90.5

TABLE VII

MIOU (IN %) FOR SEMANTIC SEGMENTATION ON THE *ben-ge-8k* DATASET [45]

Modality	U-Net [53]	SatMAE (*vit_base*)	MAPEX
RGB	56.3	53.1	**56.9**

Sentinel-2 data, while Scale-MAE [6] is trained on the RGB bands of Sentinel-2. Since Scale-MAE only provides pretrained weights for a ViT-large architecture, we use that configuration in our comparisons. Our setup is as follows: after pretraining, we retain only the top two FFN experts for each modality, pruning the others. This results in smaller, modality-specialized models (i.e., RGB experts for RGB input, SAR experts for SAR input, etc). These specialized models are also compared against the same model architecture and size (two experts) trained fully supervised from scratch (FS).

1) Specialized Models: Table V shows that FT our specialized models on their respective input modalities outperform other methods for NIR, SAR, and ELE inputs. This pattern is consistent whether we perform *k*-NN classification on features extracted from the specialized models or fine-tune the entire models. On modalities that are closer to Scale-MAE pretraining data, this model outperforms our approach. This is partly explained by the fact that Scale-MAE is 2.3× larger than our specialized models. More importantly, our method outperforms the bigger Scale-MAE on those modalities that differ the most from the pretraining data used for Scale-MAE.

Besides land-cover classification, we test MAPEX on two other specific tasks: flood and wildfire detection. These are important applications that are commonly tackled with specific remote sensing modalities (see Fig. 4).

For example, SAR (Sentinel-1) data is more informative than RGB data for detecting water in cloudy weather, while the SWIR bands of Sentinel-2 provide better information than RGB data for detecting burned areas. Since most foundation models are commonly trained on either the full spectrum of Sentinel-2 or only on RGB data, we demonstrate in Table VI that our smaller, specialized models outperform general foundation models. Specifically, for flood detection, MAPEX outperforms SatMAE by 3% (absolute) and Scale-MAE by 1%, while for wildfire detection, it surpasses SatMAE by 2% and performs comparably to ScaleMAE despite being considerably smaller. Beyond classification and detection, we also evaluate MAPEX on segmentation tasks to assess its ability to handle dense predictions. To this end, we add the same simple segmentation head to both the RGB-specific pruned model (with MAPEX) and SatMAE, and we compare the results to a U-Net baseline. Table VII shows that MAPEX achieves the best performance, demonstrating its versatility across multiple remote sensing tasks.

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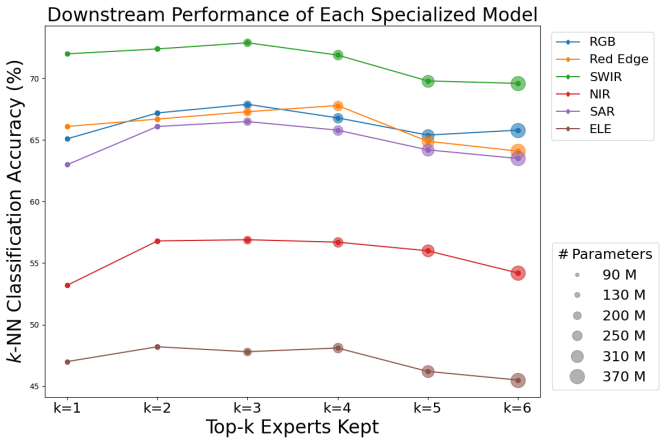


Fig. 5. Downstream accuracy with different numbers of retained modality experts k. Most models achieve peak accuracy with top-$k \in \{2, 3\}$, while more experts tend to reduce accuracy.

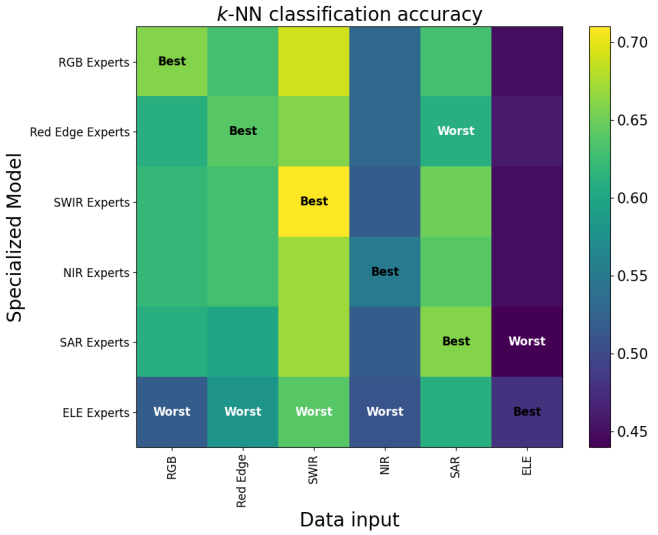


Fig. 6. Expert ↔ modality specialization on the *ben-ge-8k* dataset [45]. Each row corresponds to the performance of MAPEX pruned for a specific modality and applied across different modalities. The best and worst specialized models for each input modality are highlighted.

2) Number of Experts: We explore how the number of selected experts (top-k) influences the downstream performance of specialized models across the different modalities. We train models with up to six experts per modality and adaptively drop experts for downstream evaluation. We face a tradeoff between expert number and model size: pruned models with one expert have 90M parameters, while six experts inflate the model to 370M. Fig. 5 shows that most modality-specific models achieve their highest accuracy when using the top-2 or top-3 experts. Higher top-k values often reduce accuracy, likely because the model becomes too large, with too much capacity for a small dataset. Given that, we choose the top 2 as it yields a smaller, more efficient model with minimal loss in accuracy.

3) Specialization of Experts: This work focuses on transforming a general, modality-agnostic foundation model into smaller modality-specific models by selectively keeping experts specialized to each type of data. To test the effectiveness of these specialized models, we create modality-specific versions and evaluate them across all available data types. Results are shown in Fig. 6. This involves, for example, retaining only the RGB experts to build an RGB-specific model, then testing it with various input modalities (represented in the first row of Fig. 6). We repeat this process for each modality. The heatmap reveals that each specialized model performs best when tested on its intended data type (e.g., RGB experts on RGB data, NIR experts on NIR data). These results indicate that MAPEX expert routing successfully instills expert ↔ modality relationships into the model during pretraining.

4) Balancing Expert Size and Number: To find the ideal number of experts and sizes, we conduct an experiment with a fixed downstream model size (90M, ViT-Base) while varying the number and size of experts per modality. To ensure a fair comparison without inflating model sizes when using more experts, we fix the number of parameters added to the base model across expert configurations. Models are pretrained with different numbers of experts and pruned to a fixed number per modality: 1, 2, 5, or 10 experts are retained per modality. This corresponds to expert sizes of 4.7, 2.4, 0.9, and 0.5M parameters, respectively. Fig. 7 shows that optimal expert configurations vary by modality, but the majority of modalities achieve the highest accuracy with two experts per modality. For RGB, accuracy peaks with five smaller experts (0.9M each), suggesting that moderate specialization enhances learning. In contrast, elevation (ELE) performs best with fewer, larger experts, indicating a less diverse feature space that benefits from stronger individual experts. These findings suggest that a fixed number of experts per modality is suboptimal, but two experts per modality is a good compromise across most cases.

5) Multimodal Downstream Evaluation: Since combining multiple modalities is often beneficial for some downstream tasks, we evaluate the multimodal capabilities of MAPEX by testing its performance on different spatially co-registered data modalities: RGB, SAR, and their combination (RGB + SAR) from the *ben-ge-8k* dataset. For unimodal inputs, we use the top-2 experts per modality, while for the combined RGB + SAR input, we use the top-1 expert per modality to maintain the same model size during evaluation. Note that for the multimodal setup, we merge the data channelwise before feeding it to the model. Table VIII shows that MAPEX outperforms SatMAE and the supervised baseline for both RGB and SAR inputs while closely matching ScaleMAE, despite its significantly smaller architecture. These results show that MAPEX can easily adapt and compete with other methods, depending on the available modalities at test time.

6) Few-Shot Learning: We perform few-shot experiments to evaluate the label efficiency of our method and compare its performance with other approaches. In each k-shot experiment, we randomly select k samples for every class from the training set. We keep the same experimental setup as described in Section V-B. Table IX reports the performance of MAPEX

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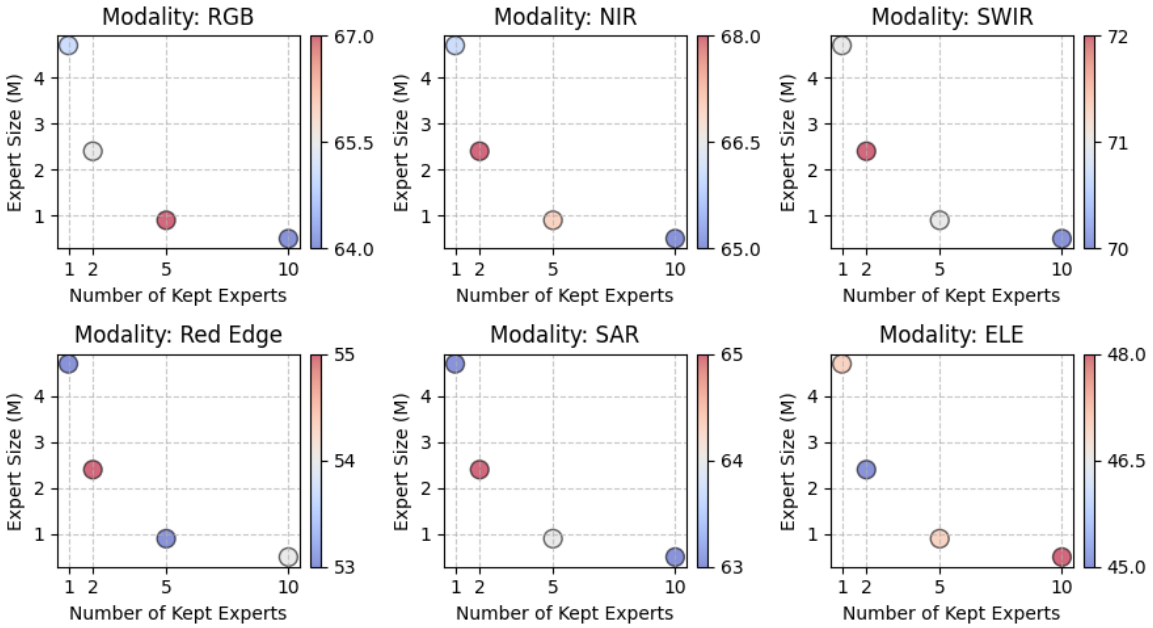


Fig. 7. Downstream accuracy (%) for different expert numbers and sizes per modality. Total model size is fixed at 90M.

TABLE VIII

k-NN AND FT PERFORMANCE (ACCURACY IN %) FOR LAND-COVER CLASSIFICATION ON DIFFERENT COMBINATIONS OF MODALITIES OF THE *ben-ge-8k* DATASET [45]

Modality	Supervised FS	SatMAE [8]		ScaleMAE [6]		MAPEX (Ours)	
		k-NN	FT	*k*-NN	FT	*k*-NN	FT
Architecture # params	*MAPEX* 130M	*vit_base* 90M		*vit_large* 310M		*MAPEX* 130M	
RGB	71.2	64.5	70.8	75.1	79.9	67.2	75.6
SAR	69.7	62.2	67.2	65.3	73.1	66.1	73.6
RGB + SAR	74.8	70.8	74.7	67.2	80.2	72.9	79.9

TABLE IX

FEW-SHOT RESULTS FOR FT (ACCURACY IN %) ON LAND-COVER CLASSIFICATION USING THE RGB MODALITY OF *ben-ge-8k* [45]

	Supervised FS	SatMAE [8]	ScaleMAE [6]	MAPEX (Ours)
Architecture # params	*MAPEX* 130M	*vit_base* 90M	*vit_large* 310M	*MAPEX* 130M
10-shot	32.1	34.5	38.9	38.6
100-shot	48.1	53.0	57.9	56.1

compared with SatMAE [8] and ScaleMAE [6] on the RGB modality of the *ben-ge-8k* dataset. The results, presented for both ten-shot and 100-shot scenarios, show that MAPEX achieves competitive performance, outperforming SatMAE in both scenarios and nearing the performance of ScaleMAE, despite ScaleMAE having a significantly larger architecture. This demonstrates that MAPEX can handle limited labeled data effectively, making it a practical and efficient alternative to larger models like ScaleMAE.

C. Computational Efficiency Analysis

Furthermore, we evaluate MAPEX from a computational efficiency perspective. Table X reports inference latency, floating point operations per second (FLOPs), and peak memory usage for MAPEX before and after pruning, as well as for SatMAE and ScaleMAE. Modality-aware pruning reduces computational cost from 866.7 to 33.9-G FLOPs and decreases peak memory from 14.8 to 0.96 GB, while improving latency by over an order of magnitude. These results demonstrate that MAPEX not only yields lightweight, modality-specialized expert models but also significantly improves runtime efficiency.

D. Implementation Details

To evaluate a pretrained model with k-NN classification, we extract and average-pool image-level features for each sample. Then, a k-NN classifier with $k = 5$ is fit to the feature representations. For FT on classification tasks, we append an

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TABLE X
COMPUTATIONAL EFFICIENCY COMPARISON OF MAPEX [BEFORE PRUNING (B.P.) AND AFTER PRUNING (A.P.)] AGAINST BASELINE MODELS. LATENCY IS MEASURED WITH BATCH SIZE 1 ON AN NVIDIA H100

Model	Params	FLOPs	Latency	Peak Memory
MAPEX (b.p.)	713M	866.7 G	187.5 ms	14.8 GB
MAPEX (a.p.)	130M	33.9 G	14.7 ms	0.96 GB
SatMAE	88M	58.2 G	6.4 ms	0.82 GB
ScaleMAE	303M	84 G	10.3 ms	1.80 GB

additional linear classification layer to the pretrained model. For pretraining and downstream evaluation, images are resized to 224×224 pixels and normalized channelwise. We fine-tune the models for 30 epochs, using AdamW [52] optimizer and a learning rate scheduler that reduces the rate when the validation loss ceases to improve.

VI. LIMITATIONS AND FUTURE WORK

This work introduced a new method to efficiently leverage large pretrained remote sensing foundation models for modality-specific downstream applications. Although many remote sensing tasks can still be addressed using data from individual, purpose-built sensors, the increasing availability of multimodal observations will drive demand for downstream models capable of jointly exploiting heterogeneous data sources. While the MAPEX approach can be applied in multimodal settings, the benefits of the proposed model-pruning strategy diminish as the number of modalities grows. This limitation suggests a need for new techniques that can more effectively manage multimodal information during pretraining, FT, and inference. Beyond multimodality, integrating multitemporal remote sensing data into this framework remains an important and open challenge for future work.

VII. DISCUSSION AND CONCLUSION

This work proposes a mixture-of-modality experts module suitable for large-scale self-supervised pretraining on multimodal data. To improve downstream performance on individual modalities, we present a pruning mechanism to derive efficient downstream models specialized for individual modalities from the pretrained model. The MAPEX model is successfully pretrained on multimodal remote sensing data, achieving good reconstruction performance across modalities. Our experiments show that the modality-based expert routing mechanism elicits specialization among expert modules, resulting in subnetworks that can be mapped to individual modalities. When pruned for a specific modality, the large-scale pretrained model turns into a much smaller specialized model that performs well on downstream tasks without any FT, as k-NN evaluation experiments show. Fine-tuned, our modality-specific models perform competitively with existing fine-tuned remote sensing foundation models.

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